Algorithmic Cosmology: Thermodynamic Traversal as Discrete State-Machine Iteration

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Abstract

Astrophysical models addressing the absolute origin and terminus of universe-state geometry operate almost exclusively on continuous thermodynamic variables, consistently trapping themselves in infinite-entropy singularity loops. This paper formally derives the Algorithmic Limit Equation (ALE), establishing universal thermodynamic evolution not as a continuous vector of radiation dissipation, but strictly as deterministic state-machine iteration bounded by maximum computable logic density. We mathematically prove that terminal topological boundaries (such as spatial origin points) function equivalently to computational logic gate resets, executing symmetrically upon satisfying established entropy accumulation thresholds without violating causal limits constraints.

Keywords: Cosmological Simulation, Thermodynamic Bounds, Universal State-Machine, Computational Singularity.

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1 Algorithmic Formalization of Thermodynamic States

Modern astrophysics defines universal expansion $S(t)$ as an unspooling field of continuous probability Shannon [1948]. However, absolute physical continuity fundamentally mandates infinite granular resolution arrays—an algorithmic impossibility in any self-stable closed domain limit.

Therefore, physical resolution must execute discretely according to standard computation limits Turing [1936].

Definition 1.1 (Universal Logic Iteration)**.** Let astrophysical realities execute bounded operation sets. The grid steps strictly forward when sub-atomic computation cycles validate:

$$\mathcal{U}_{t+1} = \Phi\left(\mathcal{U}_t, \nabla E_{thermo}\right) \tag{1}$$

2 The Entropy Boundary Limits

Thermodynamic entropy conventionally dictates an irreversible decay trajectory. However, within a strictly discrete system architecture, absolute systemic entropy forces computational limits.

Theorem 2.1 (Systemic Restart Boundary)**.** *When macro entropy averages* $\mathbf{E}[\mathcal{U}] \to H_{max}$*, structural dimensional variables mathematically exhaust processing depth. The grid inherently asserts an* $O(1)$ *deterministic state flush—conceptually modeled as expanding "Big Bang" singularity arrays Lamport [1978].*

$$\lim_{\mathbf{E} \to H_{max}} \mathcal{U}_t = C_0 \tag{2}$$

3 Conclusion

By binding universal physics explicitly to discrete iteration bounds and algorithmic logic grids, thermodynamic entropy failure is mathematically mitigated. Real structures do not drift hopelessly; they operate identically to predictable hard-coded $O(1)$ boundaries dictated by fundamental operational arrays.

References

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